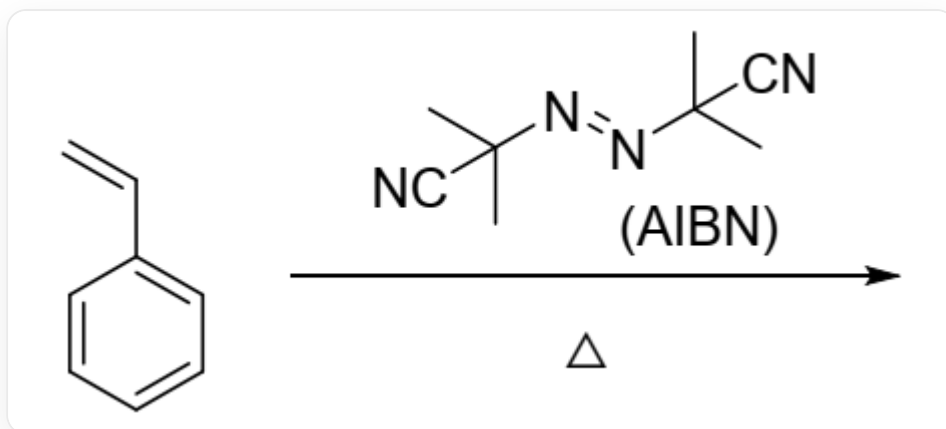


Question

Azobisisobutyronitrile (AIBN, structure as follows) can initiate the free radical polymerization of styrene to polystyrene (PS):



Styrene polymerization initiated by AIBN: `CC(C)(C#N)/N=N/C(C)(C#N)C` under heating conditions

Polymerization initiated by ^{14}C -labeled AIBN yields PS with a (number average) molecular weight of 1.30×10^6 . Scintillation counter measurements show that the AIBN radioactivity is $1.919 \times 10^8 \text{ counts} \cdot \text{min}^{-1} \cdot \text{mol}^{-1}$; the radioactivity of 5.80g PS is $858 \text{ counts} \cdot \text{min}^{-1}$.

The following statements are made:

1. AIBN heating initiates polymerization and requires the release of one molecule of gas.
2. Each molecule of the polymer has one cyano group.
3. The polymer may have a center of symmetry.
4. Each molecule of AIBN can produce two molecules of polymer.

The option below containing all correct statements is:

A. All statements are incorrect.

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 2,3,4

O. 1,3,4

P. 1,2,3,4

Answer

Correct Answer: **G**

Detailed Explanation

AIBN undergoes homolytic cleavage upon heating, generating two 2-cyano-2-propyl radicals: $\cdot\text{C}(\text{C}\equiv\text{N})\text{C}\cdot$ (the radical is located on the central carbon), and releasing one molecule of nitrogen gas.

CHECKPOINT

1 PTS

AIBN heating initiates polymerization and requires the release of one molecule of nitrogen gas, statement 1 is correct

Subsequently, the radical reacts with styrene, generating a radical at the α -position of the phenyl group to initiate polymerization.

The number of styrene monomers contained in each PS chain is: $1.30 \times 10^6 \div 104.1 = 1.25 \times 10^4$.

The number of styrene monomers corresponding to each AIBN is: $1.919 \times 10^8 \div [858 \div (5.80 \div 104.1)] = 1.24 \times 10^4$.

The two values are close; since one AIBN molecule generates two radicals, the chain termination method under experimental conditions is coupling termination. One molecule of AIBN produces one molecule of polymer.

CHECKPOINT

1 PTS

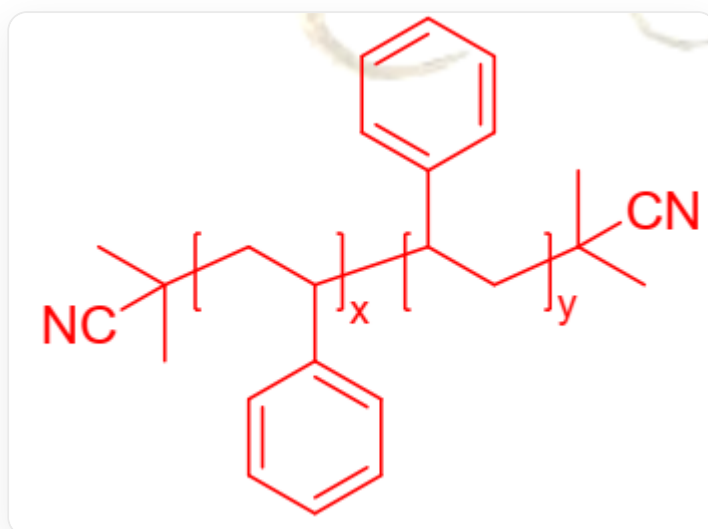
The chain termination method under experimental conditions is coupling termination. One molecule of AIBN produces one molecule of polymer, statement 4 is incorrect

CHECKPOINT

1 PTS

Calculated number of styrene monomers contained in each PS chain is 1.25×10^4 , the number of styrene monomers corresponding to each AIBN is: 1.24×10^4

Then it is easy to obtain the final polymer structure as:



The basic structure of the polymer is: CC(CC(C(C1=CC=CC=C1)CC(C)(C#N)C)C2=CC=CC=C2)(C#N)C (both degrees of polymerization x and y are 1), both repeating units are [*]CC([*])C1=CC=CC=C1 ([*] indicates that the repeating unit is bonded to other parts of the polymer)

As can be seen from the structure, each molecule of this polymer has 2 cyano groups.

CHECKPOINT

1 PTS

The molecule of this polymer has 2 cyano groups, statement 2 is incorrect

When the degrees of polymerization of the two segments are the same, i.e., $x=y$, the polymer has a center of symmetry.

CHECKPOINT

1 PTS

When the degree of polymerization $x=y$, the polymer has a center of symmetry, statement 3 is correct

Statements 1 and 3 are correct, choose G